

Market orientation, strategic flexibility and business model innovation

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Abstract

Purpose – This study aims to investigate how market orientation (MO) motivates firms to develop business model innovation and how such effects are moderated by strategic flexibility.

Design/methodology/approach – In this study, a questionnaire-based survey was undertaken to test the proposed hypotheses. The empirical study was conducted on a sample of 204 firms using two key informants (408 respondents) in China. The regression model is used to test the proposed model.

Findings – This research finds that both responsive market orientation (RMO) and proactive market orientation (PMO) have a positive effect on business model innovation. The effects of RMO and PMO on business model innovation are contingent on resource (coordination) flexibility in different ways. More importantly, this study finds that resource flexibility enhances the positive effect of RMO and weakens the positive effect of PMO. The study also finds that coordination flexibility enhances the positive effect of PMO.

Research limitations/implications – Future research can explore the internal mechanisms through which RMO and PMO promote business model innovations. Although the study finds that both the MOs promote business model innovation, they may promote business model innovation through different mediating effects. Future research can explore the role of external dynamic capabilities. This research mainly focuses on the internal dynamic capability of focal firms. However, as a focal firm-centered boundary spanning activity system, to transform into a new business model, firms not only need to reconfigure internal resource base, but also need to realign external collaboration network.

Practical implications – This research also bears important managerial implications. First, firms should be aware of the positive effect of MO on business model innovation. Firms with higher level of RMO or PMO can promote business model innovation. Second, when firms implement RMO for business model innovation, managers should focus on resource flexibility. Where MO is responsive, marketing managers need to be concerned with ensuring various applications of existing resource so as to understand effectively the current customers and market domain. Third, to leverage PMO for business model innovation, firms should adopt coordination flexibility. For firms with higher level PMO, firms should try to find the new internal coordination process for customer latent needs.

Originality/value – The conclusion extends the business model innovation research from the view of dynamic capabilities. As one of types of dynamic capabilities, MO is also the important antecedent of business model innovation. Further, this research also discusses the role strategic flexibility plays in business model innovation.

Keywords Market orientation, Business model innovation, Resource flexibility, Coordination flexibility

Paper type Research paper

Introduction

Business model innovation attracts increasing research attention because extant literature underscores the *locus* of competitive advantage has turned from focal firms, bilateral partnership to the holistic business model design (Amit and Zott, 2001; Zott and Amit, 2007; Foss and Saebi, 2017). Business model is a focal firms' boundary-

spanning activity system designed to create and capture value for stakeholders (Zott and Amit, 2010). In connected economy, firms can create competitive advantage by business model innovation (Chesbrough, 2010; Zott and Amit, 2008; Foss and Saebi, 2017). Business models are not stable over time. They frequently become subject to adaptation and innovation. Business model innovation refers to the search for a new way of doing business and design new activity system to create and capture value for its stakeholders (Zott and Amit, 2010). For example,

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enabling with digital technology, more manufacturing firms transformed from B2C business model to servitization, C2B manufacturing and internet-based intelligent manufacturing (Rabetino *et al.*, 2018).

Numerous studies have emphasized firms' motivations to engage in business model innovation. Foss and Saebi (2017) argued that neither the external antecedents nor the internal drivers of business model innovation have been sufficiently regarded. Recent literature has turned to identify the antecedents of business model innovation from rational position, cognition, learning and dynamic capability view. According to Teece (2018), the dynamic capabilities of firms are one kinds of capabilities for devising and refining business models (Teece, 2007, 2018). Following a widely accepted logic, dynamic capabilities involve sensing, seizing and transforming capabilities (Teece, 2012). Likewise, studies that focus on specific aspects of dynamic capabilities, such as opportunity recognition and flexibility, will illuminate aspects of business model innovation (Teece, 2018).

More importantly, extant literature achieved consensus that business opportunities creating or recognition is the initial step for business model innovation (Schneider, 2017). The needs of customers signal the opportunities to create values and are integral to business model innovation (Amit and Zott, 2012, 2015). Firms need to identify the consumer needs and translate these into effective value creation. Market orientation (MO) is considered to serve as sensing capabilities, which is particularly helpful in market-driven customer value creation as it helps firms to source, track, store and disseminate market information for greater awareness of business opportunities (Day and van den Bulte, 2002). A case in point is Haier, one of the first and best-known market-oriented Chinese firms. It focused on customers, competitors and other stakeholders in the market. Based on the market intelligence collected, Haier continuously meet the demands of customers (He and Wei, 2011). In 2009, Haier built a platform named Haier Open Partnership Ecosystem (HOPE). Through the HOPE, Haier implemented the responsive and proactive MO. Haier recognized the expressed needs, such as using the washing machine to wash tomato, and solving air conditioning disease. At the same time, Haier excavated the potential needs of customers, such as the designing magic mirror, which is an integration of mirror, artificial technology, health monitor and leisure in the bath room. The expressed and potential needs can facilitate the business model innovation, which is manifested in the following areas. First, Haier created the new value proposition of intelligence, eco-friendly and health. Second, Haier created new activity system including content, structure and governance. Haier adopted the new activities of software development, healthy consultancy and eco-friendly standard setting. Haier has established links with firm from different industrials, universities and R&D organizations across the world.

Opportunities creating or identifying is insufficient for business model innovation. During the innovation processes, firms must deploy resources to seize and exploit the opportunities (Teece, 2018). As transforming capabilities, strategic flexibility can reconfigure the resources to exploit the

opportunities for business model innovation. Thus, strategic flexibility, which reflects the capabilities to identify major changes in the environment, quickly commits resources to new courses of action in response to change, and acts promptly when it is time to halt or reverse such resource commitments (Katsuhiko and Hitt, 2004), may have a significant influence on the relationship between MO and business model innovation. Yet, although strategic flexibility has emerged as dynamic capabilities (Wei *et al.*, 2014), the function of strategic flexibility when realizing business model innovation has not been critically examined.

How do dynamic capabilities affect business model innovation? Despite the increasing salient of this question, researchers have mainly relied on qualitative or cases studies (Foss and Saebi, 2017; Teece, 2018). Previous research indicated that based on general dynamic capabilities, firms can realize business model innovation. However, these studies few points, out which specific types of dynamic capabilities can affect business model. Furthermore, until now, extant literature offers few specific answers to the question on how different types of dynamic capabilities jointly affect business model innovation.

Addressing these gaps, based on business model literature and dynamic capabilities theory, this research investigates the effect of RMO and PMO on business model innovation. And the moderating effect of strategic flexibility is also investigated. This research contributes to both dynamic capability literature and business model innovation literature in two ways. First, this study extends the dynamic capabilities from product innovation to business model innovation. The dynamic capabilities-based framework of MO, strategic flexibility and business model innovation is established. This study specially probes the configurations of dynamic capabilities: MO and strategic flexibility. This research views MO as sensing dynamic capabilities because it aims to discover and satisfy unmet customer needs (Slater and Narver, 1995; Atuahene-Gima, 1996; Atuahene-Gima *et al.*, 2005). As transforming capabilities, strategic flexibility may help firms to exploit the opportunities. Second, different types of dynamic capabilities play different roles in the processes of business model innovation. MO can facilitate the business model innovation, and strategic flexibility will moderate the relationship between MO and business model innovation. Therefore, to promote business model innovation, firms need to not only build sensing capability to identify opportunities, but also to build transforming capabilities which can leverage the specific types of sensing dynamic capabilities.

Theoretic background and hypothesis

Business model innovation

Business model can be defined as the activity system designed to create and capture value by exploiting business opportunity (Amit and Zott, 2001; Zott and Amit, 2010, 2013).

Business model has three core components: value proposition, activity system and resource portfolios. First, value proposition captures the dominant logics orchestrate the elements of a business model (de Oliveira *et al.*, 2018). Second, activity system is designed to center on value proposition (Palo and Tähtinen, 2011). It consists of *content*, *structure* and

governance. *Content* refers to the selection of activities, aiming to answer the question that what activities should be performed by a focal firm or other partners. *Structure* describes how the activities are linked and their relative importance for the business model. *Governance* refers to who should perform the activities and where. Third, both internal and external resource portfolios form the base of activity system because any activity is based on specific resource.

Business model innovation refers to the search for new logics of the firm and design new activity system to create and capture value for its stakeholders (Zott and Amit, 2010; Foss and Saebi, 2017). Massa and Tucci (2014) conclude that business model innovation may either refer to “the design of novel business models for newly formed organizations” or to “the reconfiguration of existing business models” (p. 424). In this study, we explore business model innovation in the context of incumbent firms where the established business model is undergoing renewal. Therefore, to promote business model innovation, firms should first identify new value proposition or design theme, and then redesign the proper activity system centering on new value proposition to capture the value. The essence of redesign activity system is the adoption of new activities (content), and/or new ways of linking the activities (structure) and/or new ways of governing the activities (governance).

Market orientation and business model innovation

Dynamic capability theory argues that, beyond rare, valuable, imperfectly imitable and non-substitutable resource base, firms should build dynamic capability to rebuild resource portfolios (Teece *et al.*, 1997). Dynamic capability is divided into sensing capability, sizing capability and also transforming capability (Teece, 2007). Sensing capability refers to the capability to search, learn, filter, shape and calibrate new opportunities (Teece, 2007; Augier and Teece, 2009). Seizing capability refers to the capacity to motivate resources for new opportunity exploitation (Teece, 2007; Wilden *et al.*, 2013). Transforming capability refers to capacity to reconfigure resources and structure (Teece, 2012).

To develop new business model, firms should identify new value proposition for customer and then translate it into specific activity system (Zott and Amit, 2010; Priem *et al.*, 2012). New value proposition often derives from organizational search for unmet customers' needs (Chesbrough, 2010). Therefore, to generate new value proposition, firms need a sensing capability that helps firms to scan, search and explore customer needs for new opportunities (Teece, 2007).

MO is defined as understanding and satisfying customers and other relevant stakeholders (Day, 1994; Narver and Slater, 1990). The research can be grouped into three major perspectives:

- 1 behavioral (Slater and Narver, 1995);
- 2 cultural (Narver *et al.*, 2004); and
- 3 capability perspective (Foley and Fahy, 2009).

MO is considered to serve as such capability that helps firms to source, track, store and disseminate market information for greater awareness of business opportunities (Day and Van den Bulte, 2002). In this paper, the capability perspective of MO is adopted that capabilities have the potential influence on

innovation. MO facilitates continuous market learning by customer and competitor analyses and disseminating process (Narver and Slater, 1990). The understanding of customer needs enables a firm to identify customer value proposition and promote business model innovation (Ramani and Kumar, 2008).

Advance in MO literature unpacks it as RMO and PMO. RMO can help firms to understand expressed unmet needs, whereas PMO helps firms to explore latent unmet needs (Narver *et al.*, 2004). RMO and PMO differ in the distribution of attention to the set of strategic issues and answers (Hurley and Hult, 1998; Wei *et al.*, 2014). First, they differ in attention distribution on customer needs. RMO attaches more attention to issues such as current customers, threats and opportunities in existing market domain (Atuahene-Gima *et al.*, 2005; Wei *et al.*, 2014). In contrast, PMO, focusing on the future or latent market, attaches more attention to issues such as future competition and new market opportunities (Wei *et al.*, 2014). Second, they differ in resource distribution pattern on customer needs. RMO guides firms to modify existing business model for better serving current customers. On the contrast, firms with PMO attempt to allocate more resources to the strategic activities beyond existing product-market domains to facilitate new experiments and discoveries (Yannopoulos *et al.*, 2012).

Strategic flexibility and business model innovation

Dynamic capability theory also argues that, to transform new value proposition into specific activity system for new business model, firms should develop transforming dynamic capability to reconfigure resource portfolios. Resource forms the base for core activity in new business model. To implement new business model, firms should reconfigure existing resource portfolio (Scott *et al.*, 2014). Strategic flexibility is a firm's capability to identify changes in the environment, to quickly commit resources to new courses of action in response to changes and to act promptly when it is time to halt or reverse such resource commitments. Extant literature suggests that strategic flexibility is one kind of transforming capability, which derives from both resource flexibility and coordination flexibility (Liu *et al.*, 2009; Wei *et al.*, 2014).

Resource flexibility and coordination flexibility differ in micro-foundation and the way to reconfigure resource portfolios. Resource flexibility derives from the width of use of existing resources. Therefore, resource flexibility increases the flexibility by creating more opinions of alternative resources in existing portfolio (Sanchez, 1995; Wei *et al.*, 2014). Coordination flexibility derives from the organizational process to create new resource combinations. Therefore, coordination flexibility enables resource reconfiguration through breaking the current structure of resource base (Sanchez, 1995; Wei *et al.*, 2014). In other words, resource flexibility is primarily concerned with the technical features of any specific resource, whereas coordination flexibility is primarily concerned with the organizational capability to reconfigure diverse resources (Li *et al.*, 2017). For example, in Haier, modular product and process architectures have been regarded as resource flexibility. This can help firms effectively perform a broad range of new tasks. The organizational process of Haier can be regarded as

coordination flexibility. This can help firms effectively deploy resource and recombine resource in new context. Therefore, according to dynamic capability theory, two types of MOs enable focal firms to generate new value proposition, whereas strategic flexibility leverages these effects by facilitating a focal firm to reconfigure resources for new business model. Based on the above discussion, we build a theoretical model as shown in Figure 1.

Hypotheses development

Market orientation and business model innovation

MO aims to meet the needs of customers and create the new value proposition. The new value proposition is the prerequisite for business model innovation. It may increase business model innovation. First, RMO may promote the development of new value proposition derived from learning from current customers. RMO focuses largely on the domain of the firm's current knowledge and experience and provides an in-depth understanding of current customers and their expressed needs. By focusing on unmet needs of current customers, firms can generate new ideas to update or modify existing value proposition.

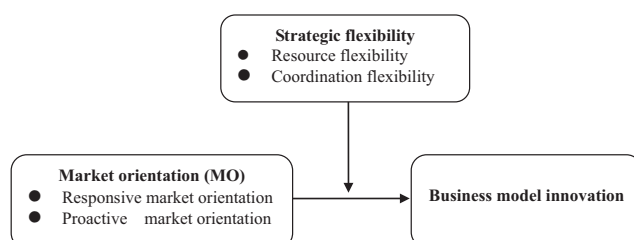
Second, PMO guide firms generate a new value proposition derived from a deep understanding of latent customer needs. PMO reflects an exploratory learning behavior involving the search for new and diverse information and knowledge that is beyond the scope of its experience (March, 1991). The market knowledge improves the understanding of current business deficiencies and provides new business opportunities. More importantly, PMO often inspires redefinition of current customer needs, which promotes exploratory value proposition (Wu *et al.*, 2013).

Third, MO may lead to the innovation of business model components and activities. The MO directs management's attention scope for gauging the potential value-added components and activities (Wei *et al.*, 2014). Firms with MO concentrate on ongoing behaviors and activities, including customer and competitor-intelligence generation. To meet the needs of customers, firms need the experiment about the content, structure and activities in the process of business model innovation (Sosna *et al.*, 2010). Through the experiment, firms can design new activity systems. Therefore, we hypothesize that:

H1a. RMO has positive effect on business model innovation.

H1b. PMO has positive effect on business model innovation.

Figure 1 Theoretic model



The moderating effect of resource flexibility

Although the RMO and PMO help firms to generate new value proposition, firms should reconfigure proper resource portfolio to transform value proposition into specific activity system. Given RMO focusing more on unmet explicit needs of current customer, it often guides firms to search for increment change of existing value proposition. Therefore, to facilitate the new value proposition, firms need to modify existing resource combinations for existing business model. Resource flexibility facilitates this transformation by offering larger room to redeploy existing resources in new use (Zhou and Wu, 2010; Narver *et al.*, 2004). With higher level in resource flexibility, the scope of resource usage can be extended and firms can have slack resource for value delivery and capture (Johnson *et al.*, 2008). When resource flexibility is higher, firms can pursue more cost-effective ways of delivering and capturing value. Therefore, we hypothesize that:

H2a. Resource flexibility strengthens the positive effect of RMO on business model innovation.

However, resource flexibility may weaken the positive effect of PMO on business model innovation. First, high level of resource flexibility prevents firms from gaining the benefits of the diversity and novelty of market information associated with PMO. With PMO, firms acquire the market information that is complex, novel and ambiguous. Hence, by focusing on latent customer needs, firms are likely to create a range of market opportunities that lie outside their experience. Pursuing such initiatives requires an adaptation of the resource combination or new way to support activities that satisfy latent and emergent customer needs. Second, resource flexibility emphasizes extensive use of extant resource which prohibits the opportunities exploitation. A high level of resource flexibility may unwittingly lead the firm into a competency or familiarity trap by which the exploitation of current customers' needs excludes newer efforts (Huang *et al.*, 2013). Hence, a high resource flexibility is likely to limit the firms with PMO to reduce the likelihood of new resource and novel information application for business model innovation. Therefore, we hypothesize that:

H2b. Resource flexibility weakens the positive effect of proactive market orientation on business model innovation.

Different from resource flexibility, coordination flexibility is based on organizational routines for dynamic resource configuration. Firms with high coordination flexibility are capable to build novel resource base. Coordination flexibility weakens the positive effect of RMO because it often disrupts ongoing activity system and reduces the consistency. High coordination flexibility is likely to diminish the effect of responsive MO on BMI because it engenders adverse conflicts that limit knowledge sharing and integration. Because responsive MO involves relatively familiar information about current customers' needs, interpreting and using familiar information effectively requires a well-defined and narrowly focused resource.

To leverage RMO for business model innovation, the need of resource reconfiguration is incremental modification of existing

activity system to quickly response to customer-expressed needs. However, the coordination flexibility is based on the organizational routines and processes to throughout reconfiguration of resource base (Sanchez, 1997). Over-reconfiguration of existing resource base incurs the disturbance in consistency of activity systems and thus high cost to transform the new value proposition to activity system. Therefore, we hypothesize that:

- H3a.* Coordination flexibility weakens the positive effect of responsive market orientation on business model innovation.

Coordination flexibility may strengthen the effect of PMO because it complements the need of resource reconfiguration derived from PMO. Value proposition derived from PMO often are novel and thus set more demand on radical reconfiguration of resource base. Firms with high coordination flexibility develop organization processes for dynamic resource configuration. Firms with high coordination flexibility can integrate, build and reconfigure resources radically with low cost and high speed time (Sanchez, 1997). Therefore, coordination flexibility creates the context in which the firm can better assimilate and apply new resources to meet latent customers' needs (Zhou and Wu, 2010), which makes firms to create easily the value proposition for the latent needs. High coordination flexibility relaxes routine inertia, which helps firms to break down its organizational routine inertia for exploration of new alternatives (Zhou and Wu, 2010). Therefore, coordination flexibility can help firms to efficiently integrate and recombine resources for the new market opportunities (Wei *et al.*, 2014). Therefore, we hypothesize that:

- H3b.* Coordination flexibility strengthens the positive effect of PMO on business model innovation.

Methodology

Sample and data collection

We conducted surveys of Chinese firms to test the hypotheses. China is one of the most important emerging economies where business model innovations are springing up. We developed questionnaire based on literatures and the interview. First, we were engaged in interviews with eight senior managers in Xi'an to collect the information on typical characteristics of environment and managerial practices. Second, we adjusted the items based on the interviews and relevant literatures. Third, we interviewed with the previous senior managers again, invited them to check the items and revised the items.

We developed an exploratory study with another 12 Chinese top managers titled as CEO, general manager and vice president. First, we promised that their response would be confidential and only applied to academic research. Second, we invited the managers to finish all the survey items. Third, we repeatedly communicated with the managers about the appropriateness of terminologies and clarity of the items. Fourth, based on their suggestions, we finally designed the questionnaire to ensure that all the items are clearly understandable. We collected data from various industries (e.g. electrical equipment, electronic

communication, chemical and pharmaceutical, general equipment, internet technology and other industries) and areas such as Shaanxi, Henan, Shandong, Guangdong and Jiangsu Provinces. We choose the companies from the firm list provided by the local governments. To ensure the access to the right respondents and accurate understanding of the items, we conducted an onsite survey. Although onsite survey is costly and time-consuming, which can ensure the questionnaire response completely and seriously (Sheng *et al.*, 2011).

To reduce potential common method bias, we invited two managers of the same company to complete different parts of the questionnaire. For each firm, we assigned two doctoral students to implement survey onsite. Through communication with managers, the doctoral students arranged the meeting time. Then they conducted the survey with Managers A and B separately, and finally checked the questionnaire after the managers finished.

In the survey, a total of 500 firms were approached and 296 firms participated. Because of missing data, our final sample includes 204 firms, which represented a response rate of 35.2 per cent. To check non-response bias, a *t*-test was performed to compare the responding and non-responding firms along firm attributes such as firm size, age and sales. All *t*-statistics were insignificant, which indicated a low possibility of non-response bias (Armstrong and Overton, 1977). Among the 408 responding managers, 64.5 per cent of them were titled as chairman of the board, CEO, general manager, vice general manager and chief engineer, and 35.5 per cent of them were titled as R&D department manager. The managers had worked for 7.55 years on the average. A total of 70.5 per cent of them had received the bachelor degree or above. Table I represents the profile of the sample.

Table I Profile of responding firms (*N* = 204)

Characteristics of responding firms	(%)
Firm age	
Less than 5 years	15.34
5-10 years	34.66
11-20 years	34.66
More than 20 years	15.34
Firm size	
Less than 50 employees	14.86
50-200 employees	36.00
200-1,000 employees	34.29
More than 1,000 employees	14.85
Firm sales (in RMB)	
Less than 10mn	15.34
10-100mn	37.71
100-1,000mn	37.71
More than 1,000mn	9.15
Firm ownership	
State-owned enterprise	13.07
Private-owned enterprise	69.32
Foreign-owned enterprise	13.06
Collectively owned enterprise	4.55
Industry	
High-tech industry	76.7
Other industries	23.3

Measures

Business model innovation

We adopted the scales developed by [Zott and Amit \(2007\)](#) to measure business model innovation. We asked top managers to indicate, on a 1–5 Likert scale, the extent to which ten statements were true regarding the business model innovation over the past three years:

- Incentives offered to participants in transactions are novel.
- The business model gives access to an unprecedented variety and number of participants and/or goods.
- The way to generate revenues is novel.
- The way to generate revenues is novel.
- Created new sources of revenues.
- Overall, the company's business model is novel.

Responsive and proactive market orientations

Following the work of [Yannopoulos et al. \(2012\)](#) and [Lamore et al. \(2013\)](#), we asked senior managers to indicate, on a 1–5 Likert scale, the extent to which nine different statements were true regarding MO in their firms over the past three years. Six of the statements pertain to RMO:

- 1 We constantly monitor our level of commitment and orientation to serving customers' needs.
- 2 Our strategy for competitive advantage is based on our understanding of needs.
- 3 We measure customer satisfaction systematically and frequently.
- 4 We are more customer-focused than our competitors.
- 5 We freely communicate information about our successful and unsuccessful customer experiences across all business functions.
- 6 Data on customer satisfaction are disseminated at all levels in this company on a regular basis.

Eight of the statements pertain to PMO:

- 1 We continuously try to discover additional needs of our customers of which they are unaware.
- 2 We search for opportunities in areas where customers have a difficult time expressing their needs.
- 3 We extrapolate key trends to gain insight into what customers will need in the future.
- 4 We help our customers anticipate developments in their use of our products and services.
- 5 We incorporate solutions to unarticulated customer needs in our products and services.
- 6 We brainstorm on how customers use our products and services.
- 7 We innovate even at the risk of making our own offering obsolete.
- 8 We work with lead users of our offerings to recognize customer' needs months in advance of the majority of users.

Resource flexibility and coordination flexibility

We adopted the scales used by the work of [Sanchez \(1995, 1997\)](#), [Zhou and Wu \(2010\)](#) and [Wei et al. \(2014\)](#). We asked senior managers to indicate, on a 1–5 Likert scale, the extent to which nine different statements were true regarding strategic

flexibility in their firms over the past three years. Five of the statements pertain to resource flexibility:

- 1 A typical material handling system can handle different part types.
- 2 Cross-trained workers can perform a broad range of manufacturing tasks effectively.
- 3 A typical machine can perform many types of operations.
- 4 A typical part can use many different routes.
- 5 We can produce a wide variety of products in our plants.

Four of the statements pertain to coordination flexibility:

- 1 The firm can redeploy the use of resource for alternative products.
- 2 The firm can reconfigure the resource chain for new products easily.
- 3 The firm is capable to adjust organizational processes effectively.
- 4 The firm is capable to create new combinations of resources effectively.

Control variables

A variety of organizational and environmental factors such as firm age, size, industry type, industry life cycle, competitive intensity, demand uncertainty and legal incompleteness are controlled for the alternative explanations for business model innovation. Firm age and firm size may affect firm growth by create inertia ([Hannan and Freeman, 1984](#)). Firm age is measured by calculating the natural logarithm of the number of years since a firm was founded, and firm size is measured by calculating the natural logarithm of the number of employees ([Rothaermel and Deeds, 2004](#)). The study also controls industry type which could determine the companies' market opportunities and the frequency of business model innovation ([Ali et al., 1993](#); [Klevornick et al., 1995](#)). Industry life cycle was assessed by the stage of the most key product in its life cycle (1 = Introduction stage, 2 = Growth stage, 3 = Maturity stage and 4 = Decline stage).

The external environmental factors, such as competitive intensity, demand uncertainty and legal incompleteness, also have effect on business model innovation. Following the study of [Wei et al. \(2017\)](#), we measured competitive intensity by a five-item scale:

- Competition in our industry is cutthroat.
- Anything that one competitor can offer, others can match readily.
- Price competition is a hallmark of our industry.
- There are many "promotion wars" in our industry.

We measured demand uncertainty by a three-item scale adapted from the work of [Jaworski and Kohli \(2003\)](#):

- 1 Customers demand is not clearly understandable.
- 2 Customer preference is difficult to predict.
- 3 Customers hold low brand loyalty.

We measured legal incompleteness with the scale from the work of [Wei et al. \(2017\)](#):

- The laws and regulations have been incomplete.
- Voids in the laws and regulations have persisted.
- The regulations and policies have not been specific enough to rely on.
- The regulations and policies have not been specific enough to guide practices.

Assessing common method variance

Following the procedures recommended by Podsakoff *et al.* (2003), this study undertook both *ex ante* and *ex post* approaches to reduce common method variance (CMV). First, because collecting data from single informant is the most important source of CMV, we asked two managers to complete the different parts of survey. One manager answered the scales for business model innovation, RMO, PMO and legal incompleteness, whereas the other answered the scale for competitive intensity, demand uncertainty, resource flexibility and coordination flexibility.

Second, to check for CMV, we applied the marker variable (MV) method (Lindell and Whitney, 2001), using a two-item variable known as government intervention as our MV, which was theoretically unrelated to at least one focal variable (Cronbach's $\alpha = 0.757$). Government intervention is measured with two five-point items:

- 1 We often have to employ some employees or collaborate with some partners according to the requirement of government.
- 2 We often have to do something out of our business according to the requirement of government.

We used the lowest positive correlation between government intervention and other focal variables ($r = 0.053$) to adjust the correlations between variables. As shown in Table IV, none of the correlations between focal variables became insignificant after MV adjustment, which indicates a reduced chance of common method bias (Lindell and Whitney, 2001).

Reliability and validity

Table II summarizes the results of reliability analysis. Table II shows that the Cronbach's alpha and composite reliability values are all greater than the 0.7 cut-off, indicating adequate reliability (Nunnally, 1978). Most of factor loadings are above 0.7 and average variance extracted (AVE) is above 0.5, which indicates a high convergent validity (Fornell and Larcker, 1981; Segars, 1997). This research also conducted confirmed factor analysis (CFA) with the items linked to prior defined constructs to test the validity. The results indicate that the measuring model fit the data well (Chi-square/df = 16.37, RMSEA = 0.045, CFI = 0.97, NFI = 0.91).

We checked the discriminant validity using Chi-square difference tests and AVE method. First, we performed ten pairwise tests with each pair of constructs. In a pairwise test, a restricted model (correlation fixed to 1) is compared with a freely estimated model (correlated estimated freely). Significant Chi-square difference is in support of discriminant validity (Anderson and Gerbing, 1988). The tests indicated that the Chi-squares of restricted models were significantly different from those of freely estimated models, which indicated high discriminant validity. Second, an examination of Table III reveals that the diagonal elements representing the square root of the average variance extracted (SRAVE) for each construct are significantly greater than the off-diagonal correlations. This satisfies Fornell and Larcker's (1981) criterion for discriminant validity. Taken together, these results show that the measures in this study possess adequate reliability and validity.

Analysis and results

Table III presents the basic descriptive statistics and correlations of the hypothesized and controlled variables. We used multivariate regression analysis to test the main effects and moderating effects (Baron and Kenny, 1986) (see Table IV). We mean-centered all the variables included in interaction terms to reduce the threat of multicollinearity (Aiken and West, 1991). The variance inflation factors in all models are below 2.1, which are well below the cut-off of 10.

Table IV reports the hierarchical regression analyses results step by step. We ran Models 1 and 2 to examine *H1* and *H2*. In the first step, we entered the control variables, which together explain a significant share of the variance (Model 1: $R^2 = 0.139$, $p < 0.05$). In the second step, we add resource flexibility and coordination flexibility, which together explain a significant share of the variance (Model 2: $R^2 = 0.362$, $p < 0.001$). In the third step, we add RMO and PMO. The results indicate that both RMO and PMO have a positive effect on business model innovation (Model 3: $\beta = 0.323$, $p < 0.001$; $\beta = 0.311$, $p < 0.01$). Therefore, *H1a* and *H1b* are both supported.

To examine the moderating effects of strategic flexibility, we ran Model 4. Model 4 shows that the coefficient of interaction between resource flexibility and RMO is significantly positive (Model 4: $\beta = 0.152$, $p < 0.01$). To gain further insight into this result, we plotted the relationships in Figure 2. As shown in Figure 2, the effect of RMO increases faster when resource flexibility increases. The result indicates that resource flexibility strengthens the positive effect of RMO on business model innovation. Therefore, *H2a* is supported.

Model 4 shows that the coefficient of interaction between resource flexibility and PMO is significantly negative (Model 4: $\beta = -0.258$, $p < 0.001$). We plotted the relationships in Figure 3. As shown in Figure 3, the positive effect of PMO on business model innovation increases more slowly when resource flexibility increases. This result indicates that resource flexibility weakens the positive effect of PMO. Therefore, *H2b* is supported. Model 4 shows that the coefficient of interaction between coordination flexibility and RMO is positive, but not significant (Model 4: $\beta = 0.038$). Therefore, *H3a* is not supported.

We will further discuss this finding in the next section. Model 4 also shows that the coefficient of interaction between coordination flexibility and PMO is significantly positive (Model 4: $\beta = 0.140$, $p < 0.01$). The summary of regression result of hypotheses is presented in Table V.

We plotted the relationships in Figure 4. As shown in Figure 4, the effect of PMO increases faster when coordination flexibility increases. The result indicates that coordination flexibility strengthens the positive effect of PMO on business model innovation. Therefore, *H3b* is supported (Figures 2–5).

Discussion

How does MO and strategic flexibility affect business model innovation? Based on dynamic capabilities theory, MO is regarded as sensing dynamic capabilities, and strategic flexibility is regarded as transforming capabilities. We find that although RMO and PMO both have positive effects on business model innovation, they need different resource flexibility or coordination flexibility for business model innovation. Especially, we find that resource flexibility enhances

Table II Reliability and validity

Constructs	Measurement items	Standardized loadings
Competition intensity	Competition in our industry is cutthroat	0.807
Alpha = 0.836	Anything which one competitor can offer, others can match readily	0.758
AVE = 0.517	Price competition is a hallmark of our industry	0.820
CR = 0.842	There are many "promotion wars" in our industry	0.735
	One hears of a new competitive move almost every day	0.793
Demand uncertainty	Customers demand is not clearly understandable	0.872
Alpha = 0.768	Customer preference is difficult to predict	0.864
AVE = 0.548	Customers hold low brand loyalty	0.742
CR = 0.779		
Legal incompleteness	The laws and regulations have been incomplete	0.733
Alpha = 0.853	Void in the laws and regulations have persisted	0.867
AVE = 0.606	The regulations and policies have not been specific enough to rely on	0.904
CR = 0.858	The regulations and policies have not been specific enough to guide practices	0.825
Responsive market orientation (RMO)	We constantly monitor our level of commitment and orientation to serving customers' needs	0.762
Alpha = 0.852		
AVE = 0.549	Our strategy for competitive advantage is based on our understanding of needs	0.738
CR = 0.879	We measure customer satisfaction systematically and frequently	0.839
	We are more customer-focused than our competitors	0.762
	We freely communicate information about our successful and unsuccessful customer experiences across all business functions	0.699
	Data on customer satisfaction are disseminated at all levels in this company on a regular basis	0.767
Proactive market orientation (PMO)	We continuously try to discover additional needs of our customers which they are unaware	0.719
Alpha = 0.892		
AVE = 0.535	We search for opportunities in areas where customers have a difficult time expressing their needs	0.761
CR = 0.900	We extrapolate key trends to gain insight into what customers will need in the future	0.815
	We help our customers anticipate developments in their use of our products and services	0.777
	We incorporate solutions to unarticulated customer needs in our products and services	0.540
	We brainstorm on how customers use our products and services	0.852
	We innovate even at the risk of making our own offering obsolete	0.825
	We work with lead users of our offerings to recognize customer' needs months in advance of the majority of users	0.799
Coordination flexibility	The firm can redeploy the use of resource for alternative products	0.860
Alpha = 0.874	The firm can reconfigure the resource chain for new products easily	0.880
AVE = 0.727	The firm is capable to adjust organizational processes effectively	0.854
CR = 0.914	The firm is capable to create new combinations of resources effectively	0.815
Resource flexibility	A typical material handling system can handle different part types	0.780
Alpha = 0.779	Cross-trained workers can perform a broad range of manufacturing tasks effectively	0.635
AVE = 0.532		
CR = 0.850	A typical machine can perform many types of operations	0.761
	A typical part can use many different routes	0.775
	We can produce a wide variety of products in our plants	0.684
Business model innovation (BMI)	The business model offers new combinations of products, services and information	0.788
Alpha = 0.896		
AVE = 0.625	Incentives offered to participants in transactions are novel	0.728
CR = 0.908	The business model links participants to transactions in novel ways	0.820
	The way to generate revenues is novel	0.855
	The business model creates new sources of revenues	0.835
	Overall, the company's business model is novel	0.842

Note: Respondents answered these questions with reference to the previous three years

Table III Descriptive statistics and correlations

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Firmage (Ln)	2.308	0.790	N/A	0.635**	-0.047	-0.203**	0.002	0.05	-0.101	0.09	0.517**	0.08	-0.002	-0.122***	-0.067	-0.236**	-0.125***	-0.021	-0.172*
2. Firm size (Ln)	5.206	1.529	0.655**	N/A	0.055	-0.239**	-0.036	0.135***	-0.167*	0.001	0.414**	0.19*	-0.055	-0.194**	-0.037	-0.114	-0.145*	0.091	-0.1
3. Food and textile	0.049	0.216	0.008	0.105	N/A	-0.159*	-0.121***	-0.22*	-0.135***	-0.333**	-0.148*	-0.126***	-0.025	-0.209**	-0.206**	-0.063	-0.067	-0.079	-0.078
4. Electronic communication	0.157	0.365	-0.139*	-0.173*	-0.098	N/A	-0.18*	-0.367**	-0.206*	0.049	-0.207**	-0.109	0.069	0.020	-0.008	0.015	0.035	-0.171*	-0.003
5. Mental and metalwork	0.069	0.253	0.055	0.019	-0.062	-0.117***	N/A	-0.252**	-0.15*	-0.15*	0.005	0.058	-0.148*	-0.043	-0.015	-0.036	-0.141***	-0.069	-0.106
6. General and special equipment	0.319	0.467	0.101	0.181**	-0.155*	-0.295**	-0.186**	N/A	-0.294**	0.007	-0.013	-0.015	-0.14***	-0.028	-0.022	-0.079	-0.041	-0.026	-0.058
7. Information technology	0.098	0.298	-0.043	-0.105	-0.075	-0.142*	-0.089	-0.225**	N/A	0.129***	-0.067	-0.044	0.06	0.125***	-0.133***	-0.017	-0.054	0.029	0.040
8. Industry (high-tech)	0.779	0.416	0.138*	0.054	-0.263**	0.099	-0.089	0.059	0.175*	N/A	-0.032	-0.049	-0.063	0.051	0.062	0.032	0.075	0.036	0.082
9. Industry life cycle	2.230	0.607	0.543**	0.445**	-0.087	-0.143*	0.058	0.040	-0.010	0.022	N/A	0.055	-0.04	-0.093	-0.106	-0.195**	-0.068	0.063	-0.104
10. Competitive intensity	3.835	0.737	0.129***	0.233**	-0.066	-0.050	0.108	0.039	0.011	0.006	0.105	0.719	0.36**	0.037	-0.068	-0.063	0.023	-0.007	-0.027
11. Demand uncertainty	2.663	0.810	0.051	0.001	0.029	0.118***	-0.087	-0.080	0.110	-0.007	0.015	0.394**	0.740	0.168*	-0.249**	-0.17*	0.058	-0.049	-0.049
12. Legal incompleteness	3.149	0.833	-0.062	-0.131***	-0.145*	0.072	0.012	0.027	0.171*	0.101	-0.035	0.088	0.212**	0.778	0.002	0.012	-0.035	-0.036	-0.008
13. Resource flexibility	3.366	0.748	-0.011	0.018	-0.142*	0.045	0.039	0.032	-0.073	0.112	-0.047	-0.011	-0.183**	0.055	0.729	0.528**	0.19**	0.129***	0.242**
14. Coordination flexibility	3.752	0.663	-0.171*	-0.055	-0.006	0.067	0.019	-0.022	0.037	0.083	-0.132***	-0.006	-0.108	0.064	0.553***	0.853	0.419**	0.294**	0.442**
15. Proactive market orientation	3.978	0.620	-0.066	-0.084	-0.010	0.086	-0.080	0.014	0.002	0.124***	-0.011	0.074	0.108	0.019	0.233***	0.450**	0.731	0.563**	0.562**
16. Responsive market orientation	3.753	0.542	0.033	0.139*	-0.022	-0.109	-0.013	0.029	0.081	0.087	0.112	0.046	-0.002	0.019	0.176*	0.331**	0.586**	0.741	0.534**
17. Business model innovation	3.630	0.680	-0.110	-0.042	-0.021	0.050	-0.047	-0.002	0.091	0.131***	-0.046	0.027	0.006	0.046	0.282***	0.472**	0.586**	0.559**	0.791
18. Government intervention (mark variable)	2.480	0.883	-0.016	0.058	0.005	-0.090	0.072	-0.003	0.082	-0.005	0.075	0.092	0.100	0.129	-0.073	-0.040	-0.004	0.053	0.133

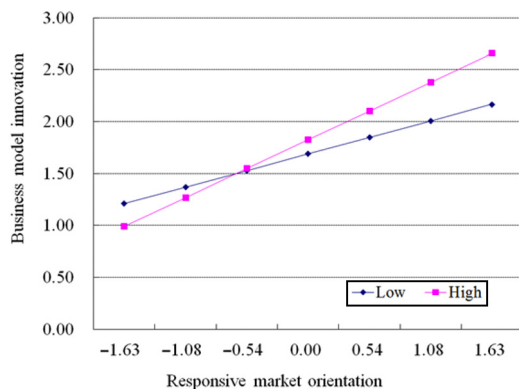
Notes: Diagonal elements in italics are square roots of the AVE values; off-diagonal elements are the correlations of the main variables of interest to the study; **** $p < 0.01$; *** $p < 0.05$; ** $p < 0.01$.

Table IV Regression results

Variables	Dependent variable: business model innovation			
	Model 1	Model 2	Model 3	Model 4
Firm age(Ln)	−0.241*	−0.068	−0.104	−0.114
Firm size (Ln)	0.006	−0.027	0.009	0.021
Food and textile	−0.062	0.060	0.036	0.023
Electronic communication	0.006	0.096	0.092	0.082
Mental and metalwork	0.029	−0.013	0.035	0.063
General and special equipment	0.020	0.096	0.063	0.039
Information technology	0.088	0.097	0.100	0.083
Industry (high-tech)	0.218**	0.092	0.032	0.043
Industry life cycle	0.188*	0.139*	0.149*	0.150*
Competitive intensity	0.118	0.045	−0.117*	−0.148*
Demand uncertainty	0.115	0.200***	0.154***	0.158**
Legal incompleteness	−0.156**	−0.145**	−0.092*	−0.110**
Resource flexibility		0.129*	0.144**	0.182***
Coordination flexibility		0.468***	0.207***	0.175**
Responsive market orientation			0.323***	0.294***
Proactive market orientation			0.311**	0.307***
Responsive market orientation × Resource flexibility				0.152**
Proactive market orientation × Resource flexibility				−0.258***
Responsive market orientation × Coordination flexibility				0.038
Proactive market orientation × Coordination flexibility				0.140**
R square	0.139	0.362	0.545	0.587
Adjusted R square	0.063	0.260	0.446	0.478
R-change		0.223	0.183	0.032
F-value	1.831*	3.568***	5.536***	5.397**
Max VIF	1.923	1.949	2.012	2.066

Notes: *** $p < 0.1$ (two-tailed test); * $p < 0.05$ (two-tailed test); ** $p < 0.01$ (two-tailed test); *** $p < 0.001$ (two-tailed test)

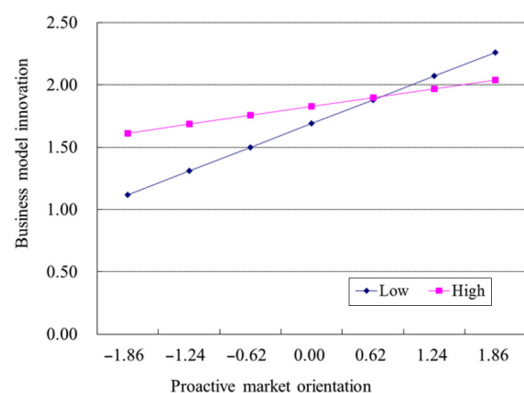
Figure 2 Resource flexibility-responsive market orientation



the positive effect of RMO while weakening that of PMO. Coordination flexibility enhances the positive effect of PMO. We find that resource flexibility negatively moderates the relationship between PMO and business model innovation. Resource flexibility can lead to core rigidities, which tend to maximize the full potential of its extant resources and ignore new value of customers from PMO. Moreover, firms are more potent in controlling its extant resources, so the costs and risks of switching the resources to grasp the new value would be higher.

Contrary to our expectation, we do not find the negative moderating effect of the coordination flexibility on the

Figure 3 Resource flexibility-proactive market orientation



relationship between RMO and business model innovation. Maybe we should further distinguish the potential and realized coordination flexibility. Although coordination flexibility can have potential to totally reconfigure the resource base, realized coordination flexibility varies with the demand of business model innovation. Business model innovation derived from RMO set more demand for resource reconfiguration and little demand for organizational reconfiguration. Therefore, coordination flexibility may not incur the cost of transforming.

Theoretical contributions

This research contributes to dynamic capability stream of business model literature in two ways. First, we link business model innovation and dynamic capability theory by investigating the alignments of different types of dynamic capabilities to promote business model innovation. Dynamic capability stream mainly agrees that sensing, seizing and transforming dynamic capabilities are needed to achieve business model innovation (Mezger, 2014; Battistella *et al.*, 2017; Teece, 2018). Different from the previous marketing research of MO and product or process innovation, this research extends the business model literature by identifying two types of MOs as antecedents of business model innovation. Although extant literature achieved consensus that sensing the existence of unmet customer needs is often the initial step for business model innovation (Teece, 2018), extant literature does not investigate the specific antecedents that promote business model innovation by sensing customer needs. This research finds that both RMO and PMO have positive effects. The findings show that sensing the explicit or latent customer needs can help firms to develop new value proposition and then promote business model innovation. This supports the view of Chesbrough (2010) that MO can overcome the cognitive constraints to realize business model innovation.

Second, this paper tests the joint effects of different types of dynamic capabilities on promoting business model innovation. We find that although RMO and PMO as sensing dynamic capabilities have positive effects of business model innovations, they need different types of transforming dynamic capabilities. Resource flexibility can leverage RMO, whereas coordination flexibility can leverage PMO to promote business model innovation. RMO often promotes business model innovation by identifying opportunities to incrementally change business model to meet the explicit need of current customer. In this case, firms often prefer to increase the internal consistency of existing business model for quick response to explicit customer needs. Therefore, resource flexibility enables firms to updating existing resource base quickly with low cost. On the contrast, PMO often promotes business model innovation by identifying novel value proposition for unmet latent needs. In this case, to deliver novel value proposition, firms often need to wholly reconfigure rather than incrementally modifying existing resource base. Resource flexibility may lead firms to search in existing resource base and thus incur competence trap. Coordination flexibility can enable firms search distantly for new resources and radically reconfigure existing resource base for new business model.

Managerial implications

This research also bears important managerial implications. First, to promote business model innovation, firms should invest in MO

Figure 4 Coordination flexibility-responsive market orientation

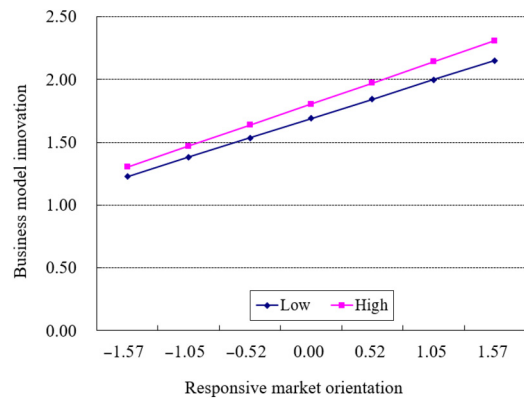
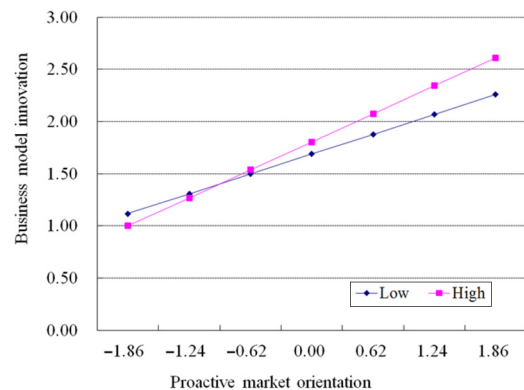


Figure 5 Coordination flexibility-proactive market orientation



to sense the unmet customer needs. Enabled with new generation of digital technology, an increasing number of firms are searching new opportunities to create new business model. We find that RMO and PMO have significantly positive effects on business model innovation. As sensing dynamic capability, two MOs can promote business model innovation by identifying new ways to do business. The unmet need of customers guides firms to generate new solution and value proposition, which guides firms develop new business model in dynamic environment. Therefore, our findings indicate that to promote business model innovation, firms should develop MO to reposition the way to serve or lead customer.

Second, firms should build proper transforming dynamic capabilities to leverage two types of MOs for business model innovation. We find that two types of MOs need different kind of transforming capabilities for promoting business model

Table V Results of hypotheses

Hypotheses	Results
H1a: RMO has positive effect on business model innovation	Supported
H1b: PMO has positive effect on business model innovation	Supported
H2a: Resource flexibility strengthens the positive effect of RMO on business model innovation	Supported
H2b: Resource flexibility weakens the positive effect of PMO on business model innovation	Supported
H3a: Coordination flexibility weakens the positive effect of RMO on business model innovation	Not supported
H3b: Coordination flexibility strengthens the positive effect of PMO on business model innovation	Supported

innovation. To leverage RMO, firms should invest more on resource flexibility. On the contrast, to leverage PMO for business model innovation, firms should invest more in coordination flexibility rather than resource flexibility.

Limitation and future directions

Despite its contributions, this study also has limitations that should be addressed in future research. First, future research can explore the internal mechanisms through which RMO and PMO promote business model innovations. Although we find that both the MOs promote business model innovation, they may promote business model innovation through different mediating effects. The two MOs may also promote different types of business model innovations. This research does not investigate this point. Addressing on this point can extend our understanding on the underlying mechanisms or micro-foundation of sensing dynamic capabilities for business model innovation.

Second, future research can explore the role of external dynamic capabilities. This research mainly focuses on the internal dynamic capability of focal firms. However, as a focal firm-centered boundary spanning activity system, to transform into a new business model, firms not only need to reconfigure internal resource base, but also need to realign external collaboration network. Therefore, firms may need to build external dynamic capability (e.g. alliance reconfiguration capability, external stakeholder engagement capability) to facilitate this transformation process.

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